English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Applied Mathematics / Mathematics, 31 May 2026 Test 2 _____ *The exact grading formula will depend on the results. As an approximation, use 2 + the number of points. Every answer must be justified and supported by clear, detailed calculations. Electronic devices (except standard calculators) and additional materials are prohibited. Time allowed: 4 hours. Good luck!*

Problem 1. (1.5 points)

1. (0.5 points) Let the continuous random vector (X, Y) have joint density

$$f_{X,Y}(x,y) = \begin{cases} cxy, & 0 < y < \sqrt{x} < 1, \\ 0, & \text{otherwise.} \end{cases}$$

   Find c, $\mathbb{E}[X]$, $\mathbb{E}[X \mid Y = 1/3]$ and $\mathbb{P}(X - Y < 1/2)$.

2. (0.25 points) Let X, Y be random variables. Prove that $\operatorname{Var}(X) = \mathbb{E}[\operatorname{Var}(X \mid Y)] + \operatorname{Var}(\mathbb{E}[X \mid Y])$.[1]

3. (0.25 points) Let $X \sim \mathrm{N}(0,1)$ and $Y = X^2$. Find $\operatorname{Corr}(X, Y)$.

4. (0.5 points) Let X_n have density $f(x) = nx^{n-1}\mathbb{1}_{(0,1)}(x)$. Show that X_n converges in probability. What about $n(1 - X_n)$?

Problem 2. (1.25 points) Two independent systems monitor a server for incidents. The system load at a given time is $\Theta \sim \mathrm{Unif}(0,1)$ percent. The first monitoring system is new and depends on the load: its detection time satisfies $X \mid \Theta = \theta \sim \mathrm{Exp}(\theta)$. The second is old and calibrated: it depends on neither the load nor the first monitoring system, and its detection time is $Y \sim \mathrm{Exp}(1)$.

1. (0.25 points) What is the expected detection time of the first system?

2. (0.25 points) Find $\mathbb{P}(X < Y)$, the probability that the new system detects the incident before the old one.

3. (0.5 points) Find the conditional density of $\Theta \mid X < Y$.

4. (0.25 points) Compare $\mathbb{E}[\Theta \mid X < Y]$ and $\mathbb{E}[\Theta \mid Y < X]$.

Problem 3. (1.25 points) Choose X uniformly in $[0,1]$. Let $X_1, X_2, \ldots$ be the digits in its decimal expansion,[2] so $X = 0.\overline{X_1 X_2 X_3 \cdots}$.

1. (0.5 points) What is the distribution of X_n? Are the digits independent? Justify your answers.

2. (0.25 points) Cantor's grandson simulates numbers in $[0,1]$ satisfying his condition: every digit occurs at least once and, for some $k \in \{0, 1, \ldots\}$, $X_{k+1} = X_{k+2} = X_{k+3} = 2$. What is the probability of drawing such a number?

For a digit d, let $Y_d^{(n)}$ be the proportion of the first n digits of X that equal d, and define

$$Y_d = \begin{cases} \lim\limits_{n\to\infty} Y_d^{(n)}, & \text{if the limit exists,} \\ 0, & \text{otherwise.} \end{cases}$$

3. (0.25 points) Find the distribution of Y_d for each $d = 0, 1, \ldots, 9$.

4. (0.25 points) Calculate $\mathbb{P}(0.08 < Y_d^{(10000)} < 0.12)$. How many digits must we observe to claim that $|Y_d^{(n)} - Y_d| \le 0.01$ with 95% confidence?

Problem 4. (1 point) Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim} \mathrm{Unif}(-\theta, \theta)$, where $\theta > 0$.

1. (0.75 points) Find the maximum likelihood estimator of θ. Determine whether it is unbiased and consistent.

2. (0.25 points) Find a method of moments estimator of θ.

Problem 5. (Bonus, 1 point) Choose three points independently and uniformly on a circle of radius 1. Find the probability that the longest side L of the triangle they form is no longer than $\sqrt{3}$. Consider a generalization: how can one calculate $\mathbb{P}(L < \varepsilon)$ for $\varepsilon \le \sqrt{3}$?

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[1] Conditional variance is defined by $\operatorname{Var}(X \mid Y) = \mathbb{E}[(X - \mathbb{E}[X \mid Y])^2 \mid Y]$.

[2] You may assume that the digits are well defined, since almost every number in $[0,1]$ has a unique representation in base r, for $r \ge 2$.